

23. Determine the IDFT of $Y(k) = \{1, 0, 1, 0\}$

$$\mathcal{DFT}^{-1}[Y(k)] = y(n) = \frac{1}{4} \sum_{k=0}^3 Y(k) e^{j \frac{2\pi kn}{4}} ; \text{ for } n = 0, 1, 2, 3$$

$$\therefore y(n) = \frac{1}{4} \left[Y(0)e^0 + Y(1)e^{j \frac{\pi n}{2}} + Y(2)e^{j \pi n} + Y(3)e^{j \frac{3\pi n}{2}} \right]$$

For integer n ,
 $\sin \pi n = 0$.

$$= \frac{1}{4} [1 + 0 + e^{j \pi n} + 0] = \frac{1}{4} [1 + \cos \pi n + j \sin \pi n] = 0.25(1 + \cos \pi n) ; \text{ for } n = 0, 1, 2, 3$$

When $n = 0$; $y(0) = 0.25 (1 + \cos 0) = 0.5$

When $n = 1$; $y(1) = 0.25 (1 + \cos \pi) = 0$

When $n = 2$; $y(2) = 0.25 (1 + \cos 2\pi) = 0.5$

When $n = 3$; $y(3) = 0.25 (1 + \cos 3\pi) = 0$

\ $y(n) = \{0.5, 0, 0.5, 0\}$

15. Is $h(n) = (-1/4) \delta(n+1) + (1/2) \delta(n) - (1/4) \delta(n-1)$ is stable and causal? Justify. (May 2016)
LTI system is stable if and only if its impulse response is absolutely summable . ie.,

$$\sum_{K=-\infty}^{\infty} (h(k)) = 0 < \infty$$

Therefore above system is Stable. This system depends on future samples. Therefore it is non-causal system.

2. What is frequency warping in Bilinear transformation? (May 2016/ April 2018)

The mapping of frequency from Ω to ω is approximately linear for small value of Ω & ω . For the higher frequencies, however the relation between Ω & ω becomes highly non-linear. This introduces the distortion in the frequency scale of digital filter relative to analog filter. This effect is known as warping effect.

Chebyshev Filter

1. The Magnitude response of Chebyshev filter will not decrease monotonically with frequency because it exhibits ripples in pass band or stop band.
2. The Transition width is very small
3. For the same specifications the order of the filter is small and is less complex and inexpensive.
4. The poles of Chebyshev filter lies along the ellipse.
5. Magnitude response produces ripples in the pass band or stop band thus it is known as equiripple filter.

Basis for Comparison	FIR Filter	IIR Filter
Stands for	Finite Impulse Response	Infinite Impulse Response
Nature	Non-recursive	Recursive
Computational Efficiency	Less	Comparatively more
Usage	Difficult	Quite easy
Feedback	Absent	Present

What is phase delay and group delay in FIR filters?



The phase delay () of the filter is **the amount of time delay each frequency component of the signal suffers in going through the filter**. 2. The group delay () is the average time delay the composite signal suffers at each frequency.

2.What is the difference between fixed point arithmetic and floating point arithmetic?

(April

2018/Nov2019)

Fixed point arithmetic	Floating point arithmetic
Fast operation	Slow operation
Small dynamic range	Increased dynamic range
Relatively economical	More expensive due to costlier hardware
Round-off errors occur only in addition	Round-off errors can occur with both multiplication and addition
Overflow occurs in addition	Overflow does not arise

11. What do you understand by input quantization error? (Nov2013/Nov2019)

In DSP, continuous time input signals are converted into digital using a 'b' bit ADC. The representation of continuous signals amplitude by a produces an error known as input quantization error.



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லிஸ்ட் தி அப்ப்ளிகாடிஸ்



DSP applications include **audio and speech processing, sonar, radar and other sensor array processing, spectral density estimation, statistical signal processing, digital image processing, data compression, video coding, audio coding, image compression, signal processing for telecommunications, control systems, ...**



define circular buffering in dsp



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dsp இல் வட்ட இடையக?



Circular buffering is **an efficient method of storing the input data of a real-time system**. Employing this technique, we need to perform only a single memory write operation for each new sample. With a GPP, we may have to implement the circular buffer in software. 13-Nov-2017